

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPE-602: Elementary Nanoscience and Exotic Materials

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** parts out of the following: [4 × 3.5 = 14]

- (a) Discuss the Fibonacci sequences and their relevance to quasi-crystals.
- (b) Explain briefly the main features of Colossal Magnetoresistance (CMR).
- (c) How is focusing of electrons performed in a transmission electron microscope (TEM)? Name the two lenses used in a TEM.
- (d) In which characterization technique are secondary electrons exploited? Discuss the interaction of electrons with thin samples.
- (e) Define Quantum dot, quantum wire, and quantum well with suitable examples.

Question 2.

- (a) What do you understand by tunneling? Discuss the salient features of scanning probe microscope (SPM) and write the construction and working principle of scanning tunneling microscope (STM). [4 + 10 = 14]

Question 3.

- (a) What are the essential differences between crystals and quasi-crystals? Compare them in a tabulated form. Why does one need a six-index system in order to index quasi-crystalline diffraction patterns? [14]

Question 4.

- (a) Discuss the structure, properties, and applications of carbon nanotubes (CNTs). Discuss in detail the synthesis of carbon-based nanomaterials using DC arc discharge techniques. [9]
- (b) Write a short note on the arc-discharge method for synthesizing single-walled carbon nanotubes (SWNTs). [5]

OR

- (a) Discuss with an appropriate diagram the mechanism of Giant Magnetoresistance (GMR). [10]

- (b) Explain the spin valve effect. [4]
-

Question 5.

- (a) What are superconductors? How are they different from perfect conductors? Explain the Meissner effect and supercurrents in superconductors. [14]

OR

- (a) Discuss the salient features of High-Temperature Superconductors (HTS). Draw a neat diagram of the structure of Y-123 ($\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$) and point out the Cu–O plane in which charge carriers exist. [14]